

- 2 Use the substitution $z = x + y$ to find the solution of the differential equation

$$\frac{dy}{dx} = \frac{1 + 3x + 3y}{3x + 3y - 1}$$

for which $y = 0$ when $x = 1$. Give your answer in the form $a \ln(x+y) + b(x-y) + c = 0$, where a , b and c are constants to be determined. [7]

[illegible]